Abdelrahman Elsayed

Electronics & Communications Engineer

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Engineer with industry and laboratory experience in embedded sensing, cellular communications, and industrial systems integration. Builds prototypes and tests how devices, networks, and software work together.

Engineering Experience

Solutions Serafin Inc. | Wireless Systems Specialist, EV Connectivity | Jan 2024 - Oct 2026

- Connected an EV motor testbed to the ICPS Lab using MQTT for maintenance telemetry and analysis.
- Built a 5G QoS simulation to evaluate delivery of urgent maintenance messages during network congestion.

ICPS Lab, Polytechnique Montréal | Graduate Researcher | 2024 - Oct 2026

- Developed Python and C++ simulation workflows, validated Siemens PLC interfaces, and investigated lab connectivity faults.

Egyptian Space Agency | Electronics and Communications Engineering Intern | Jun - Oct 2021

- Built a Raspberry Pi prototype to model onboard computing and telemetry for satellite swarms.

ENPPI | Telecommunications and Instrumentation Intern | Jun - Jul 2019

- Supported work involving optical fiber, SCADA, and instrumentation wiring.

Selected Projects

Cloud-native 5G platform | Built a local Kubernetes and Helm deployment with service probes, isolated data networks, and documented recovery tests.

5G Standalone protocol validation | Built a repeatable lab with Open5GS and UERANSIM and traced controlled faults using Wireshark, Linux state, and Python tooling.

Wearable sensing | Contributed to embedded sensor acquisition and data preparation for a collaborative monitoring and classification prototype.

Education

M.A.Sc., Industrial Engineering | Polytechnique Montréal | 2026

B.Sc., Electronics and Communications Engineering | Egypt-Japan University of Science and Technology | 2022

Technical Skills

Electronics and communications: Raspberry Pi, ESP32, I2C, SDR, 5G NR, LTE, QoS, BLE, Wireshark
Software and integration: Python, C++, Bash, Linux, ns-3/5G-LENA, Kubernetes, Helm, MQTT, Siemens PLC

Research and Teaching

Coauthored a journal article on 5G QoS for EV maintenance and three wearable-sensing conference papers. Publications
Co-lectured in fault analysis and maintenance at Polytechnique Montréal and supported electronics and communications labs at E-JUST.